

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Noise pollution - Set 1

Mark scheme

AQA spec 3.4.3.2.13; Pollution; textbook pp. 286-295

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • The human ear is sensitive to a very wide range of sound pressures / powers (about 10^{12} range) (1) • A logarithmic scale compresses this into a convenient range of numbers: each 10 dB increase represents a ten-fold increase in sound intensity (1)	2
01.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Difference = 30 dB = 3 x 10 dB (1) • = 10^3 = 1000 times (1)	2
01.3	<i>1 mark for each point.</i> • 88 dB (1) • Doubling the sound intensity increases the level by 3 dB (logarithmic scale) (1)	2
01.4	<i>1 mark.</i> • 0 dB is set as the quietest sound that can be heard (at 1000 Hz) - the threshold of human hearing; quieter sounds exist but cannot be heard (1)	1
01.5	<i>1 mark for each point.</i> • 20 to 20 000 Hz (1) • dB(A) weights sounds between about 1000 and 4000 Hz more heavily (1) • Because the nerve cells / hair cells in the cochlea are more sensitive to these frequencies so they are heard more easily / cause more annoyance; range narrows with age or noise damage (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-287; AQA spec 3.4.3.2.13.	

Question	Marking guidance	Marks
02.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Stress-related illness: ulcers, high blood pressure, heart disease (1) • Behavioural changes: irritability, aggression, lack of concentration leading to accidents (1) • Communication problems / educational underperformance (eg schools under flight paths) (1) • Sleep disturbance (1)	3
02.2	<i>1 mark for each point.</i> • Acoustic fatigue: objects exposed to sound at their natural resonant frequency vibrate, causing stress cracking / structural failure (eg early jet engine cowlings) (1) • Vibration damage: repetitive vibration from heavy vehicles damages buildings, bridges, underground pipes (1) • Shock impacts: the force of a sudden very loud noise (sonic boom) breaks windows / dislodges masonry (1)	3
02.3	<i>1 mark for each point.</i> • Birds that sing to establish territories / attract mates cannot be heard against background noise (1) • Birds scared by sudden noise leave nests so predators take eggs / chicks; cliff-nesting seabirds knock eggs or chicks into the sea (1)	2
02.4	<i>1 mark for each point.</i> • Whales / dolphins use hearing / echolocation to find food and communicate; loud sonar causes hearing damage if close to the source (1) • Behavioural changes / disorientation possibly linked to mass strandings; reduced feeding success (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 287; AQA spec 3.4.3.2.13.	
03.1	<i>1 mark for each point.</i> • A turbofan at the front forces bypass air around the inner cowl / core jet (1) • The bypass air smooths the flow of the turbulent hot exhaust gases, reducing the turbulence that creates noise (1) • The higher the ratio of bypass air to jet exhaust the quieter the engine (Trent 1000 quietest, Olympus noisiest) (1)	3
03.2	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Chevron (serrated) nozzles mix bypass air with surrounding air more smoothly (1) • Acoustic liners in the engine cowlings absorb noise (1) • Hush kits fitted to older low-bypass engines (1) • Fairings on undercarriage and wing-flap hinges reduce turbulence; welded rather than riveted panels (1) • Lighter composite aircraft need less thrust (1) • Blended wing with engines on top so the body shields the ground (1)	2

Question	Marking guidance	Marks
03.3	<i>1 mark for each point.</i> • Constant (3°) descent avoids periods of high thrust in a stepped descent and keeps aircraft higher for longer (1) • Night flights (especially take-offs, which are noisier) are restricted / banned when people are sleeping and background noise is low (1) • Noisier aircraft pay higher charges / use up more quota points and fines for exceeding limits, so airlines are encouraged to buy quieter aircraft (1) • More time to slow down after landing so wheel brakes can be used instead of noisy reverse thrusters (1)	4
03.4	<i>1 mark.</i> • Engines run at higher power / maximum thrust and the steep climb angle needs full power (although the aircraft climbs quickly out of earshot) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 288-292; AQA spec 3.4.3.2.13.	
04.1	<i>1 mark for each point.</i> • Wheel vibration: track polishing / grinding, sound-absorbing ballast, sound-absorbing suspension (1) • Pantograph: aerodynamic fairing (1) • Wheel squeal: lubrication of wheels / track on corners (1) • Braking: composite-material brakes instead of cast iron (1)	4
04.2	<i>1 mark.</i> • Roads pass through / near almost all residential areas and traffic is continuous through the day, unlike airports or railways which affect limited areas (1)	1
04.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Sound-absorbing road surfaces (porous asphalt / rubber-crumb asphalt) (1) • Acoustic fences, embankments / baffle mounds, walls or dense tree planting (1) • Traffic management for free flow (less acceleration noise); re-routing traffic away from residential areas (1) • Double glazing / acoustic insulation of buildings (1) • Vehicle design: better aerodynamics, engine insulation, quieter exhausts, quieter tyres (1)	3
04.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Air bubble curtains around the pile driver absorb noise (1) • Acoustic survey to detect cetaceans first and delay work if present; increase volume gradually (soft start) so animals can leave (1) • Drilling instead of pile-driving (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 292-294; AQA spec 3.4.3.2.13.	

Question	Marking guidance	Marks
05.1	<i>1 mark for each point.</i> • Use a calibrated sound meter (dB(A)) at fixed distances (eg 10, 50, 100, 250, 500 m) from the track behind each barrier and at a control site with no barrier (1) • Standardise the noise source: same cars / same race or a standard test noise; same time of day, wind speed / direction, weather; same height of meter above ground (1) • Take repeated readings at each distance and calculate means (and standard deviation) to assess variability (1) • Use a measure such as LA10 or Leq over a set period rather than a single peak reading (1) • Compare reduction in dB relative to the control at each distance; plot noise level against distance for each barrier; statistical test (t-test) to assess significance (1)	5
05.2	<i>1 mark.</i> • Wind carries sound / affects how far it travels so different wind directions would change readings independently of the barrier (1)	1
05.3	<i>1 mark for each point.</i> • LA10 is the noise level exceeded for 10% of the period - a measure of the noisiest periods (1) • LA90 is the level exceeded for 90% of the period - a measure of the quietest / background level (1)	2
05.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Double / triple glazing (1) • Acoustic insulation in walls / roof (1) • Bedrooms on the side facing away from the track (1) • Dense hedges / trees and earth mounds in gardens (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 293-295, 418; AQA spec 3.4.3.2.13.	

Question	Marking guidance	Marks
06.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Principles: dB is logarithmic ($10\text{ dB} = 10\times$; $3\text{ dB} = 2\times$) so small reductions in dB are large reductions in intensity; effects on humans (hearing, stress, behaviour) and wildlife (breeding birds, bats, cetaceans) • Aircraft design: high bypass ratio engines, chevron nozzles, hush kits, acoustic liners, fairings, composites, blended wing • Aircraft operation: take-off angle, flight paths avoiding urban areas, constant descent, reduced reverse thrust, night restrictions, Chapter noise limits, fines and noise charges, quota count • Airport design / location: away from population centres, taxi and engine-test areas away from housing, acoustic walls, baffle mounds, land-use restrictions, free acoustic insulation • Railway: track polishing, sound-absorbing ballast and suspension, pantograph fairings, wheel lubrication, composite brakes • Road: porous / rubber asphalt, acoustic fences, embankments, tree planting, aerodynamics, engine insulation, traffic management and routing, double glazing • Industrial: silencers on compressors, pressing instead of stamping, nylon rollers, drilling instead of pile-driving, bubble curtains, timing of blasting, acoustic mats / curtains, remote operation, ear protection • Domestic: acoustic absorbers in appliances, volume limiters, selection of quieter appliances, ear defenders • Evaluation: which approaches are most effective / cost-effective; conflicts (airports near cities for convenience); legislation and monitoring (Leq 57 dB contours) needed to enforce	9
	Model answer - Level 3 (9 marks)	
	Because the decibel scale is logarithmic, a 10 dB cut is a tenfold reduction in intensity and 3 dB halves it, so modest-sounding design changes matter. Aircraft engines have been made quieter by high-bypass-ratio turbofans, chevron nozzles, hush kits and acoustic liners, with fairings and composite airframes reducing airframe noise. Operation can be changed too: steeper take-off angles and flight paths that avoid housing, constant-descent approaches, restricted reverse thrust, night curfews, Chapter noise limits enforced by fines and quota counts. Airport location and layout - away from population centres, engine-test areas and taxiways kept from housing, acoustic walls and baffle mounds, land-use planning and free insulation for residents - deal with the noise that remains.	

Question	Marking guidance	Marks
	Railways use track polishing, sound-absorbing ballast and suspension, pantograph fairings, wheel lubrication on curves and composite brakes. Roads use porous or rubber-crumb asphalt, acoustic fences and embankments, tree planting, engine insulation and aerodynamics, traffic routing and double glazing. Industry fits silencers to compressors, presses rather than stamps, uses nylon rollers, drills instead of pile-driving, surrounds piling with bubble curtains to protect cetaceans, times blasting, hangs acoustic mats and operates remotely, with ear protection as a last line. Domestic appliances use acoustic absorbers and volume limiters.	
	Design changes at source are the most effective because they reduce noise for everyone; operational limits depend on enforcement; location works only for new developments, and airports are still sited near cities for convenience. Legislation and monitoring of noise contours (57 dB L _{eq}) are what turn these options into actual reductions.	
	Weaker answer - Level 2 (5 marks)	
	Noise pollution can be reduced in different ways. Aircraft can be designed with quieter engines and can fly at night less often. Airports can be built away from cities. Roads can have sound barriers next to them and quieter road surfaces. Trains can have their tracks polished and use better brakes. Factories can use quieter machines and workers can wear ear defenders. Noise can also be reduced by planting trees. These methods reduce the noise that reaches people.	
	What makes the difference	
	• Level 2 gives correct but generic examples ('quieter engines'); Level 3 names the design features (high bypass ratio, chevron nozzles, acoustic liners) the question's 'named examples' requires.	
	• Level 3 organises by design / operation / location as the question asks; Level 2 mixes them.	
	• The decibel-scale point and the closing evaluation of which approaches work best lift Level 3 into the top band.	
06.2	<i>1 mark.</i> • The area within which the average (equivalent continuous) sound level between 07:00 and 23:00 in summer exceeds 57 dB - taken as the onset of significant community annoyance (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-295; AQA spec 3.4.3.2.13.	